



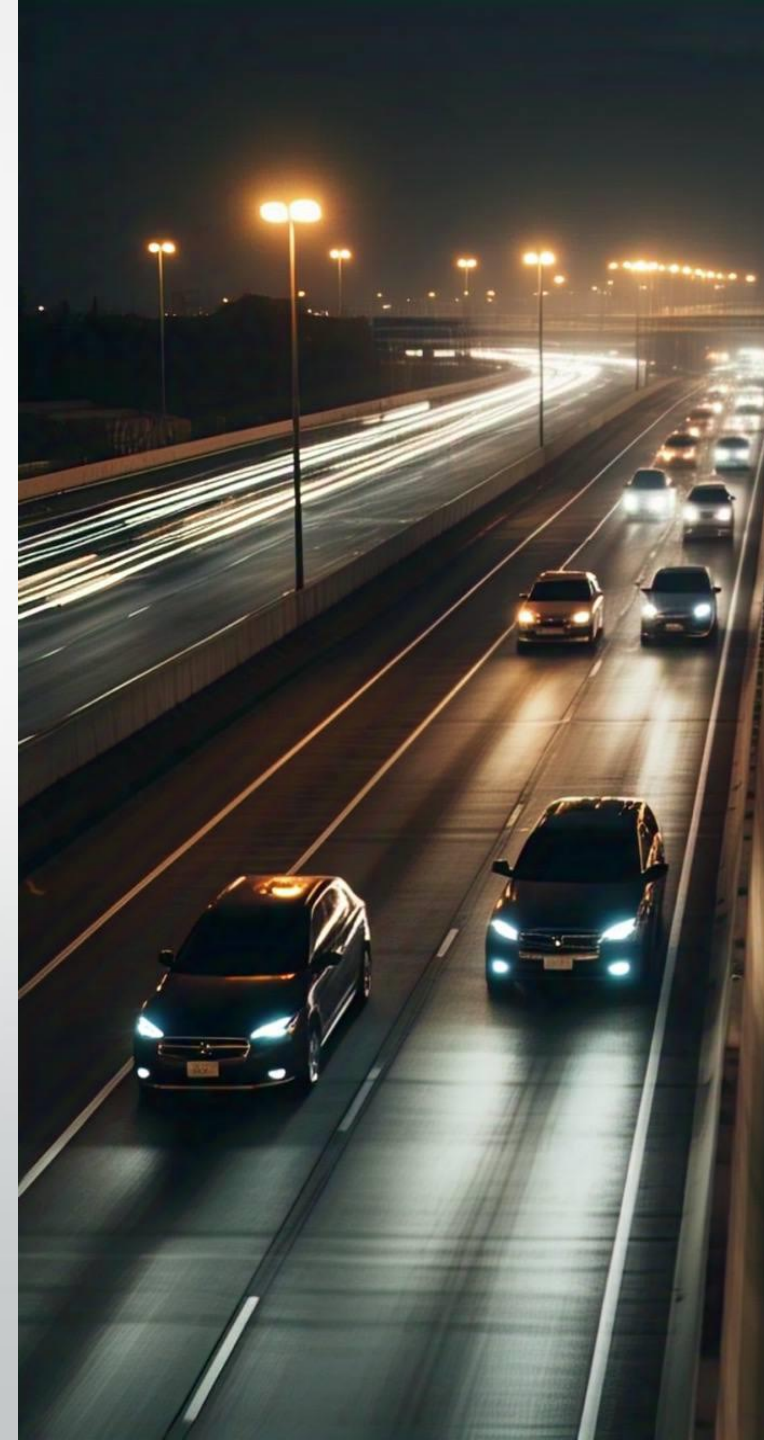
AUTOMATIC HEADLIGHT DIMMER USING ARDUINO

GROUP MEMBERS

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OBJECTIVE

- Design and implement an automatic headlight dimmer for vehicles. The aim is to reduce glare for oncoming drivers and improve visibility. We will create a cost-effective and easily adaptable system.



INTRODUCTION

High beam headlights cause glare, which can temporarily blind oncoming drivers and lead to accidents, especially at night. To address this issue, an automated system using an LDR (Light Dependent Resistor) detects approaching vehicle headlights and dims the high beams accordingly. This solution enhances road safety, eliminates the need for manual headlight adjustments, and is both cost-effective and easy to implement in vehicles.

Instant Reaction	Safety First	Market Demand
Reacts to light changes immediately	Reduces glare for safer driving	ADAS market is rapidly growing

METHODOLOGY

The automatic headlight dimmer system works by detecting the brightness of an approaching vehicle's headlights using an LDR (Light Dependent Resistor). The sensor sends data to the Arduino microcontroller, which processes the input and determines whether the headlights need to be dimmed. If an oncoming vehicle is detected, the Arduino sends a PWM (Pulse Width Modulation) signal to a relay module, which adjusts the headlight brightness by switching from high beam to low beam. Once the vehicle passes, the system restores the high beam automatically

- **LDR** - Detect ambient light levels
- **ARDUINO** - Analyze sensor data
- **PWM Control** - Adjust headlight brightness
- **Relay Switching** - Switch between high and low beams

COMPONENTS REQUIRED & BUDJET

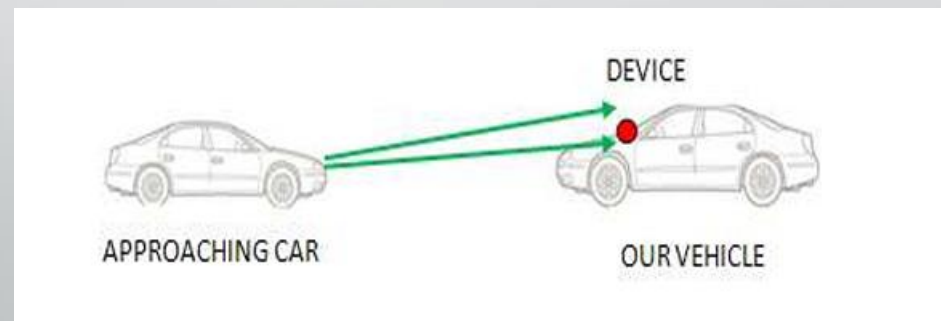
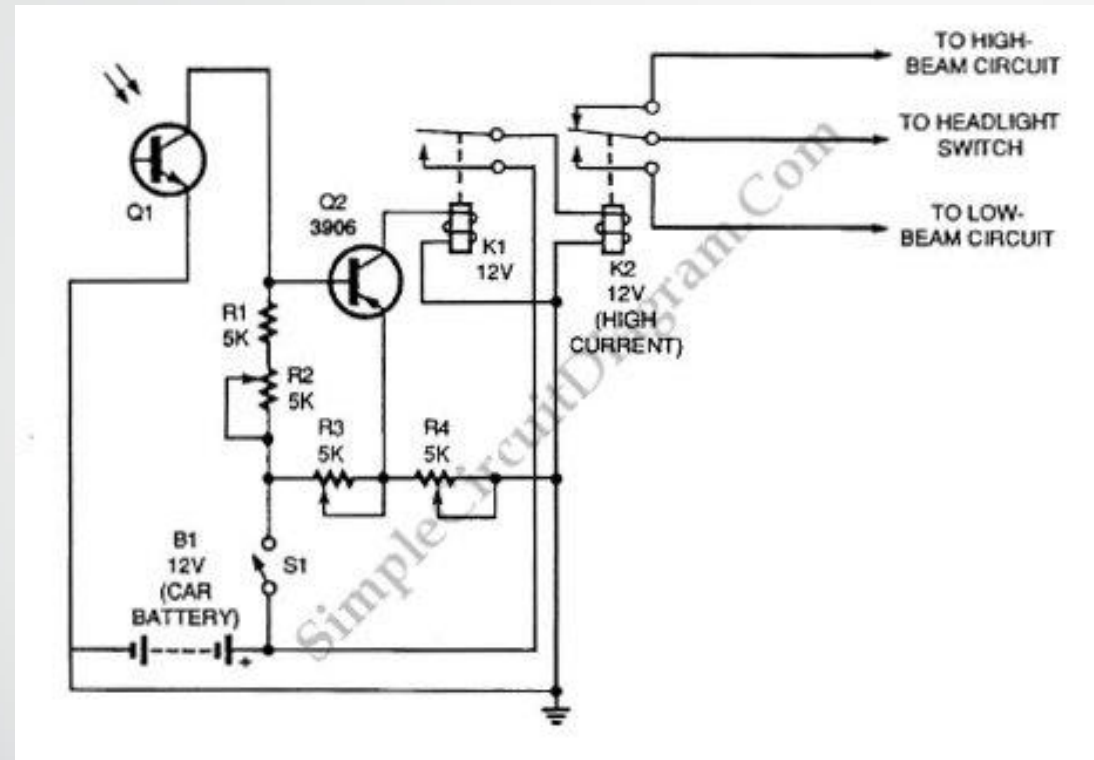
COMPONENTS	APPROX.PRICE (INR)
ARDUINO UNO	600 - 1000
LDR SENSOR	200 - 300
RESISTOR	100 - 200
MOSFET	200 - 300
RELAY	50 - 100

❑ These are the components used in this project , approximate cost of this project is 2000 – 2500 INR

PROJECT TIMELINE

- ✓ WEEK 1 : RESEARCH & DISCUSSIONS FIND PROJECT
- ✓ WEEK 2 : COMPONENTS PROCUREMENT
- ✓ WEEK 3 : CIRCUIT ASSEMBLING & TESTING
- ✓ WEEK 4 : DEBUGGING AND TESTING
- ✓ WEEK 5 : SUBMITTING THE PROJECT TO RESPECTED AUTHORITIES

CIRCUIT DAIGRAM



CONCLUSION

The automatic headlight dimmer enhances road safety. It reduces glare for oncoming drivers by automatically adjusting the headlight brightness based on ambient light conditions. This project is a cost-effective solution.

- Enhanced Safety Reduces glare for oncoming drivers.
 - Cost-Effective Uses readily available components.
 - Automatic Operation Adjusts headlight brightness automatically
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